Reader: In any case, if a function is continuous everywhere in the domain (has no discontinuity points), its graph is a continuous line: it can be drawn without lifting the pencil from the paper.

Author: I agree. I would like to emphasize that the continuity of a function at a point x guarantees that a very small displacement from this point will result in a very small change in the value of the function.

Let us turn to Fig. 27, which is the graph of the function $y=\{x\}$. Consider, for instance, $x=0.5$. The function is continuous at this point. It is quite evident that at a very small displacement from the point (either to the left or to the right) the value of the function will also change only a little. Quite a different situation is observed if $x=1$ (at one of the discontinuity points). At $x=1$, the function assumes the value $y=0$. But an infinitesimal shift to the left from the point $x=1$ (take, for example, $x=0.999$, or $x=0.9999$, or any other point no matter how close to $x=1$) will bring a sharp change in the value of the function, from $y=0$ to $y\approx 1$.

Reader: Quite clear. I must admit, however, that the local nature of the concept of a continuous function (i.e. the fact that the continuity of a function is always related to a specific point x) does not quite conform to the conventional idea of continuity. Because continuity typically implies a process and, consequently, a sort of an interval. It seems that continuity should be related not to a specific moment of time, but to an interval of time.

Author: It is an interesting observation. This local character is a manifestation of one of the specific features of calculus. When analyzing a function at a given point x, you used to speak about its value only at this specific point; but calculus operates not only with the value of a function at a point but also with the limit of the function (or its absence) at this point, with the continuity of the function at the point. It means that on the basis of the information about a function at a given point we may construct an image of the behaviour of the function in the vicinity of this point. Thus we can predict the behaviour of the function if the point is slightly shifted from x.